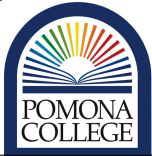


concordance=TRUE

	Environmental Analysis Teaching and Research Laboratory	Date: 7/12/2015	SOP No. 32 v0.1
	Standard Operating Procedure	Title: Soil Loss on Ignition	
	Approved By: Los Huertos	Revised: June 13, 2025	

1. Scope and Application

- 1.1** Loss on Ignition (LOI) is used to determine organic matter content (OM%) in soil samples. This is important for indirectly estimating soil organic carbon (SOC) in agricultural topsoil.
- 1.2** The applications implement SFU Soil Science Lab method (Robertson, 2011) *and add Aaron's method paper for temperature*.

2. Summary of Method

- 2.1** This method will cover how to conduct a LOI procedure in the Pomona College Geology Department Lab. Ignition of the soil samples in a muffled furnace burns off any organic matter and water content, leaving only the mineral content. The weight difference of the samples before and after ignition represents the OM%.

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3. Biases and Interferences

3.1 The following are parameters that might influence the results of the LOI test and thus, we review methods that would best fit the type of soils from our samples.

Clay content Soil texture has many influences, two of which being climate and parent material

Ignition temperature If the temperature is not hot enough, there may be OM remaining in the samples after ignition.

Ignition duration If the samples are not burned for long enough, there may be OM remaining in the samples after ignition.

4. Health and Safety

4.1 The muffled furnace burns up to 3000°C, posing a significant threat to 4th degree burns if precautions aren't taken.

Safety and Personnel Protective Equipment

4.2 Care should be taken to prevent burns. Remove crucibles from the furnace with tongs and heat-proof gloves. Wait at least 1 hour for them to cool down before handling with bare hands. In case of contact, run skin under cold water for 15 minutes and follow lab procedures for emergency.

5. Personnel & Training Responsibilities

5.1 Researchers training is required before the procedures in this method can be used.

5.2 Researchers using this SOP should be trained for the following SOPs:

- SOP01 Laboratory Safety
- SOP07 Using Balances
- SOP15 Using Drying Ovens

6. Required Materials and Apparati

1. Porcelain crucibles
2. Soil samples
3. Muffled furnace
4. Drying oven
5. Tongs
6. Heat-proof gloves
7. Jack stand
8. Metal tin
9. Combustion residue mapping sheet

7. Reagents and Standards

8. Estimated Time

8.1 This procedure requires several days to complete.

9. Sample Collection, Preservation, and Storage

10. Procedure

Preperation

10.1 Prepare the soil samples in advance. Samples should be sieved with a 2mm sieve. Use a drying oven to remove moisture content from the soil and the crucibles (optional) that may affect the true dry weights.

1. Weigh out more sample mass than desired on an metal tin, make sure to label (etch with pencil) and tare out weight of the tin.
2. Oven-dry tins at 105°C for 24 hours. The oven can fit X samples.
3. Take samples out of the oven using tongs or gloves after oven-drying. Cool them in a desiccator.
4. (Optional) Oven-dry crucibles in the same manner. Weigh them once finished while still warm.

Ignition

10.2 Once ready for ignition, we can begin heating the oven while preparing the crucibles. The Geology Dept. has multiple muffled furnaces. This SOP instructs how to use the Nabertherm furnace and load it while hot.

10.3 Use the dial knob to set the furnace to the correct program preset called "LOI" and assure it is set to the correct temperature of 360°C. Press the start button. The LCD screen will tell you the current temperature inside the furnace.

10.4 Then, for each sample...

1. Record the weight of the crucible on a scale capable of precision to 0.0001
2. Weigh out 5.0000g 0.0001g of the oven-dried sample into a crucible.
3. Record this as the pre-ignition weight.

10.5 As you prepare the crucibles, use the Combustion Residue Mapping Sheet to map out assign them Sample IDs, and correspondingly where you'll put each sample in the furnace. This is important to prevent mixing up samples as any marker or pencil written on the crucibles will burn off. Use the hole on the plate to help you determine direction.

10.6 Once the furnace is ready, use the tongs and heat-proof gloves to place the samples according to your map.

10.7 Set a timer for 2 hours

10.8 Find something to do in the meantime like prepping the next batch of crucibles.

Determining Proportion of Soil Passing No. 10 Sieve

10.9 Soils are defined as particles that are less than 2 mm. Thus, we begin by removing the gravel, particles that exceed the sand-sized particles, >2.0 mm. These particles, however, influence the soil texture, so we will quantify the contribution of particles that exceed 2 mm.

10.10 We can do this by analyzing the soil that can pass through the No. 10 sieve, which has a mesh size of 2 mm.

Measuring and Correcting for Soil Water Content

10.11 We will use air-dried soil for our test. However, even air-dried soil contains water. This moisture is associated with clay particles, salts, and organic matter and is often referred to as hygroscopic water. To calculate the soil moisture content, we weigh the soil before and after oven drying the soil at 110° C. Once we determine the moisture content, we can use the moisture content to correct for the water in the air-dried soil. Because this method relies on the change in soil mass, the water content is often referred to as gravimeter water content as opposed to a volumetric water content.

10.12 This temperature is somewhat arbitrary, and clay minerals in particular may contain 10-15% water (dry basis) at 400° C (Gardner 1986). As temperature increases, first water in soil pores evaporates, then water adsorbed to mineral surfaces, followed by water between lattice layers and that which forms part of the mineral lattice itself. The exact quantities and patterns of release in a heterogeneous mixture like soil depends on the particular mix of minerals making up a sample. Water adsorbed to organic components (as well as other volatile organic substances) will also evaporate over a range of temperatures. The key point is to specify the temperature used when reporting moisture data.

10.13 To determine the hygroscopic water content,

1. Weigh and record the weight of a soil tin (Appendix A, Box 13) and the tin # (Appendix A, Box 12);
2. Remove a 10-15 gram sub-sample from the air-dried soil that passed through the No. 10 sieve, put the sub-sample into the pre-weighed soil tin and record the weight (Appendix, Box 14);
3. Dry the sample in a drying oven set at 110 ° C for approximately 24 hours;
4. Allow the tins to cool and record the oven-dried weight (Appendix, Box 15).

10.14 We calculate the hygroscopic moisture content we subtract the mass of oven-dried sample and the air-dried mass before drying, usually less than one, unless there is no hygroscopic moisture, then is is one. Use the following equation to calculate the water content

$$\text{Moisture}_H = (\text{air-dry mass w/tin} - \text{oven-dried mass w/tin}) / (\text{air-dried mass} - \text{tin mass}) \quad (1)$$

Moisture_H can be calculated using R code developed for this SOP. See Appendix B for a user guide for the program.

Calculating Soil Sample Mass

10.15 We then calculate the “Total Sample Represented” in the hydrometer test as the mass of soil in hydrometer test as the oven dry mass used by the percent passed through 2 mm sieve and multiply by 100.

10.16 We will analyze air dried soil – that has passed through a No. 10 sieve.

$$W = \text{ovendrymass} * \text{percentpassing2mmsieve} * 100 \quad (2)$$

$$WB = \frac{M_{air} * 10000}{Per_{No.10} * (100 + \text{Moisture}_H)} \quad (3)$$

Obs.	Soil Reading Times	Temperture	Blank Hydrometer
0	Before Inverting (0 seconds)	X	X
1	40 seconds (Repeat twice)		
2	2.0 minutes		
3	5.0 minutes	X	X
4	15 minutes		
5	30 minutes		
6	60 minutes	X	X
7	2 hours	X	X
8	3 hours (optional)	X	X
9	6 or 8 hours (optional)	X	X
10	24 hours	X	X

Table 1: Reading times modified from **citetstandard2007d422**. These times can be adjusted, be sure to record the actual times in Boxes 19 and 20.

10.17 But since the sample has hygroscopic water, we will use the correction factor above to weigh a “oven-dried equivalent”, DW_e .

10.18 If the soils are coarse textured ($\geq 70\%$ sand), will will use DW_e of 100 g, otherwise 50 g is sufficient.

10.19 Since air dried soil contains water, the soil sample used will weigh slightly more than 50g or 100 g for sandy soils where $DW_e = DW_{air} * (1 + DW_c)$ ¹, thus we can re-arrange the equation to determine the amount of air-dried soil to use for our test:

$$DW_{air} = 50 / (1 + DW_c) \quad (4)$$

and for sandy soils,

$$DW_{air} = 100 / (1 + DW_c) \quad (5)$$

Be sure that the cylinder is back from the edge of the counter and in a location where it won't be disturbed.

Determining Proportion of Soil Retained No. 200 Sieve

10.20 Particles retained on the No. 200 sieve are larger are often categorized as sand. As the hydrometer does very little to distinsihin between size fractionation of sand – very fine, fine, medium, course, and very course sand.

10.21 But collecting the particles on the No. 200 sieve, we can use seives to better understand the soil characteristics. We can do this by analyzing the mass of the soils retained on the No. 200 sieve, which has a mesh size of XX 0.125 mm.

¹I need to check this!

1. Pour the contents of the cylinder through a No. 200 sieve;
2. Using the sink with a sediment trap, wash the soil in the sieve until the water leaving the sieve is clear.
3. Select a soil tin, and record the tin # (Box XX) and tin mass (Box XX).
4. Carefully wash the soil from the tin into the soil, without damaging the sieve screen. The screens are quite delicate, so using a squeeze bottle is probably best. Using a brush or metal device will force particles into the screen, which can get stuck, distort and stretch the mesh openings or otherwise damage them.
5. Weigh the dry soil with the soil tin and record the results in Box Xx.

11. Recording the Results

Record the results in Appendix A, Then enter the data into the data entry form (<http://test>). Other columns calculations will be produced with the R code from Appendix B.

Do not put commas in text fields – if you do the download... it will create extra columns and the data will be misaligned.

12. Data Analysis and Calculations

Once the data has been entered in the online form, you can extract the data as a csv file. This csv file can then be called by R to calculate the data automatically, using the following functions.

Importing Data

Import and check data integrity

```
#hydrometer = import()
#head(hydrometer)
#str(hydrometer)
```

What should you be looking for? We want to make sure the data shown are the same as expected. There are several ways to look at the data to determine this. I have shown two ways. 1) Looking at the header (first six observations), then we evaluate the structure of the dataframe.

Calculating effective diameter, D_e

Calculating the Percent Finer, PF

Graphing and Interpreting the Results

13. QC/QA Criteria

14. References

14.1 D422...

14.2 Wilde, S.A., R.B. Corey, J.G. Iyer, and G.K. Voigt. 1979. Soil and Plant Analysis for Tree Culture. Oxford and IBH Publishing Co., New Delhi. Pp. 12-13.

14.3 Anderson, J.M., and J.S. Ingram, eds. 1993. Tropical Soil Biology and Fertility: A Handbook of Methods. 2nd ed. CAB International, Wallingford, UK. P. 93-94.

15. Appendix A. Data Sheet

Grain Size Analysis–Hydrometer Datasheet

1. Project		2. Researcher:		3. Date	
4. Sample ID		5. Hydrometer Number/Type			
Sample Preparation					
6. Pre-Treatment		☐ HCl		☐ H ₂ O ₂	
				☐ Fe–O	
7. Beaker ID		8. Dispersing Agent		9. Specific Gravity of Particles $GS_p =$	
				10. Desired Sample ☐ 50 g ☐ 100 g	
11. Total Soil for Split (g)			12. Soil Passing Sieve No. 10 (g)		
13. Tin #:		14. Tin Tare Weight		15. Tin w/Air-Dried Soil	
				16. Tin w/Oven-Dried Soil	
Calculated Values					
17. % Passing No. 10		18. % Hygroscopic Correction Factor		19. Effective Soil Weight WS_e	
				20. Actual Air-Dried Soil Weight WS_a	

Hydrometer Readings and Calculations *K is a constant that is calculated by the temperature and density of the suspension.

21. Time	22. Elapsed Time (t) sec/min	23. Actual Hydrometer Reading (R_a)	24. Blank Hydrometer Reading (R_b)	25. Temp. °C	26. K constant*	27. Effective Depth (L)	28. Particle Diameter (D_e)	29. PF Partial	30. PF Total

No. 200 Sieve Processing

31. Tin #	32. Tin Mass (g)	33. Dry Soil w/Tin (g)
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Quality Control/Quality Assurance

34. Researcher (Signature)	35. Data Entry By (Signature)	36. Quality Check By (Signature)
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16. Appendix B: R code to calculate parameters for the SOP